

CENG381 Stochastic Processes

Stationary Distribution — Practice 1 (Answer Key)

Q1 Solution

a) Transition matrix:

From A: $P(A \rightarrow A) = 1 - 2/3 = 1/3$, $P(A \rightarrow B) = 2/3$

From B: $P(B \rightarrow A) = 1/5$, $P(B \rightarrow B) = 4/5$

	A	B
A	$[1/3 \quad 2/3]$	
B	$[1/5 \quad 4/5]$	

b) Stationary distribution (balance equations):

$$\begin{aligned}\pi_A &= \pi_A \cdot (1/3) + \pi_B \cdot (1/5) & \dots \text{ (I)} \\ \pi_B &= \pi_A \cdot (2/3) + \pi_B \cdot (4/5) & \dots \text{ (II)} \\ \pi_A + \pi_B &= 1 & \dots \text{ (III)}\end{aligned}$$

From (I): $\pi_A - (1/3)\pi_A = (1/5)\pi_B$

$$(2/3)\pi_A = (1/5)\pi_B$$

$$\pi_B = (10/3)\pi_A$$

Substitute into (III):

$$\pi_A + (10/3)\pi_A = 1 \rightarrow (13/3)\pi_A = 1$$

$$\pi_A = 3/13 \approx 0.231, \quad \pi_B = 10/13 \approx 0.769$$

c) Interpretation:

$\pi_A = 3/13 \approx 23.1\%$. In the long run, a tagged fish spends about 23% of its days in area A and 77% in area B. This asymmetry arises because the rate of leaving A (2/3) is much higher than the rate of leaving B (1/5).

Q2 Solution

a) Balance equations $\pi \cdot P = \pi$:

$$\begin{aligned}\pi_W &= \pi_W \cdot (1/2) + \pi_M \cdot (1/2) + \pi_F \cdot 1 & \dots \text{ (I)} \\ \pi_M &= \pi_W \cdot (1/3) + \pi_M \cdot 0 + \pi_F \cdot 0 & \dots \text{ (II)} \\ \pi_F &= \pi_W \cdot (1/6) + \pi_M \cdot (1/2) + \pi_F \cdot 0 & \dots \text{ (III)} \\ \pi_W + \pi_M + \pi_F &= 1 & \dots \text{ (IV)}\end{aligned}$$

From (II): $\pi_M = (1/3)\pi_W$

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$$\begin{aligned}\text{From (III): } \pi_F &= (1/6)\pi_W + (1/2) \cdot (1/3)\pi_W \\ &= (1/6)\pi_W + (1/6)\pi_W = (1/3)\pi_W\end{aligned}$$

Substitute into (IV):

$$\pi_W + (1/3)\pi_W + (1/3)\pi_W = 1 \rightarrow (5/3)\pi_W = 1$$

$$\pi_W = 3/5 = 0.6, \quad \pi_M = 1/5 = 0.2, \quad \pi_F = 1/5 = 0.2$$

b) Verification $\pi \cdot P = \pi$:

$$\begin{aligned}(\pi \cdot P)[W] &= 0.6 \cdot (1/2) + 0.2 \cdot (1/2) + 0.2 \cdot 1 = 0.30 + 0.10 + 0.20 = 0.60 = \pi_W \quad \checkmark \\ (\pi \cdot P)[M] &= 0.6 \cdot (1/3) + 0.2 \cdot 0 + 0.2 \cdot 0 = 0.20 = \pi_M \quad \checkmark \\ (\pi \cdot P)[F] &= 0.6 \cdot (1/6) + 0.2 \cdot (1/2) + 0.2 \cdot 0 = 0.10 + 0.10 = 0.20 = \pi_F \quad \checkmark\end{aligned}$$

c) Detailed balance test:

$$\begin{aligned}\text{Pair (W, M): } \pi_W \cdot P_{\{WM\}} &= 0.6 \cdot (1/3) = 0.20 \\ \pi_M \cdot P_{\{MW\}} &= 0.2 \cdot (1/2) = 0.10 \\ 0.20 &\neq 0.10 \rightarrow \text{detailed balance FAILS for (W,M).} \\ \text{Pair (W, F): } \pi_W \cdot P_{\{WF\}} &= 0.6 \cdot (1/6) = 0.10 \\ \pi_F \cdot P_{\{FW\}} &= 0.2 \cdot 1 = 0.20 \\ 0.10 &\neq 0.20 \rightarrow \text{detailed balance FAILS for (W,F).}\end{aligned}$$

Conclusion: the chain is NOT time-reversible. There is a net directional flow $W \rightarrow M \rightarrow F \rightarrow W$, which is characteristic of a non-reversible process.

Q3 Solution

a) Transition matrix P (order F, N, L):

	F	N	L
F	[1/2	1/2	0]
N	[1/4	1/2	1/4]
L	[0	1	0]

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b) Balance equations $\pi \cdot P = \pi$:

$$\begin{aligned}\pi_F &= (1/2)\pi_F + (1/4)\pi_N && \dots \text{ (I)} \\ \pi_N &= (1/2)\pi_F + (1/2)\pi_N + \pi_L && \dots \text{ (II)} \\ \pi_L &= (1/4)\pi_N && \dots \text{ (III)} \\ \pi_F + \pi_N + \pi_L &= 1 && \dots \text{ (IV)}\end{aligned}$$

From (I): $(1/2)\pi_F = (1/4)\pi_N \rightarrow \pi_N = 2\pi_F$

From (III): $\pi_L = (1/4) \cdot 2\pi_F = (1/2)\pi_F$

Substitute into (IV):

$$\pi_F + 2\pi_F + (1/2)\pi_F = 1 \rightarrow (7/2)\pi_F = 1$$

$$\pi_F = 2/7 \approx 0.286, \quad \pi_N = 4/7 \approx 0.571, \quad \pi_L = 1/7 \approx 0.143$$

Verification of (II): $(1/2) \cdot (2/7) + (1/2) \cdot (4/7) + (1) \cdot (1/7) = 1/7 + 2/7 + 1/7 = 4/7 = \pi_N \checkmark$

c) Interpretation:

The ATM is in the Low state on about $1/7 \approx 14.3\%$ of days.

$\pi_F = 2/7 \approx 28.6\%$ (Full), $\pi_L = 1/7 \approx 14.3\%$ (Low).

So the ATM is Full about twice as often as Low.

The Normal state ($\pi_N = 4/7 \approx 57.1\%$) is the most frequently visited.